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High-temperature wettability to slag of carbon nanotube modified MgO castable prepared by a chemical vapor deposition method

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Abstract

Conventional MgO castable can not meet the requirements of steel production due to its poor slag corrosion resistance. Thus, in this work, a 1 wt% carbon nanotubes modified MgO (1 wt%-CNTs/MgO) castable was successfully prepared through the catalytic pyrolysis of polyethylene (PE) powder *via* a chemical vapor deposition strategy, the effects of pyrolysis temperature, catalyst concentration and type, and PE powder dosage on the preparation of the modified castable samples were investigated. Among the catalysts tested, Ni showed the best performance, and the optimal preparation conditions for the 1 wt%-CNTs/MgO castable were a Ni concentration of 1.0 mol/L, a pyrolysis temperature of 973 K and a sample:PE powder mass ratio of 1:1. Moreover, the as-prepared 1 wt% CNTs/MgO sample exhibited a super-nonwetting property toward slag with a contact angle of 116° at 1723 K, and the corrosion depth of the sample was only 1092 μm, which was approximately 46% lower than that of the control MgO castable sample.

Previous

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Keywords

PE powder; CNTs/MgO castable; High-temperature wettability; Slag corrosion resistance

1. Introduction

MgO-based castables exhibit excellent high-temperature mechanical properties and a simple production process, making them widely used in the steelmaking industry [[1], [2], [3], [4], [5], [6], [7], [8]]. However, their slag corrosion resistance still needs further improvement. Common nano-carbon additives in refractories include CNTs, graphene, and carbon black. However, the hydrophobic nature of these carbon sources increases water demand during castable preparation. Carbon nanotubes (CNTs) were introduced in to shaped MgO-C refractories to improve their high-temperature service performances [[9], [10], [11], [12]]. The addition of nano-carbon materials can fill pores and then enhance the physical properties of the final product. For example, Valipour A et al. [13] prepared low carbon magnesia-carbon refractories using multi-walled carbon nanotubes (MWCNTs) as the carbon source, which reduce the graphite content from 20 wt% to 3 wt% in the final product, and the residual strength after thermal shock of the sample reached 93%, higher than the 89% of the original sample. Luo et al. [14] prepared Al_2O_3-C refractories using MWCNTs as the carbon source, and the results demonstrated that the specimens containing 0.05 wt% MWCNTs exhibited enhanced mechanical properties. Nevertheless, the properties of the prepared refractory samples deteriorated with the further increase of the MWCNTs amount from 0.1 to 1 wt%, which was attributed to the agglomeration of MWCNTs and their negative effects on the ceramic phase morphology. Moreover, direct incorporation of CNTs into refractory suffers from the problem of high cost.

Using carbon-containing gases produced by the pyrolysis of waste plastics as carbon sources has proven to be a practical way to prepare CNTs at low cost [[15], [16], [17], [18], [19], [20], [21], [22], [23]]. For instance, Li et al. [24] investigated the preparation of CNTs from the pyrolysis of polyethylene using a series of Fe/ZSM-5 (zeolites, microporous and mesoporous materials) catalysts with different Fe loading (5, 10, 20 and 30 wt%), and the results showed that the highest CNTs production (262.24 mg/g_{PE}) was obtained using the 20Fe/ZSM-5 catalyst. Li et al. [25] studied the production of CNTs from the catalytic pyrolysis of waste polyethylene (PE), polypropylene, polystyrene terephthalate, and polyethylene over a Ni/Al_2O_3 catalyst, indicating that PE waste plastic was more suitable as feedstock for the preparation of high-quality CNTs by the catalytic pyrolysis strategy. Our group has prepared CNTs with yields up to 59.0 wt% from waste PE using low-cost cobalt nitrate as a catalyst precursor [26], subsequently, graphene/CNTs aerogels with excellent hydrophobicity and mechanical stability were successfully synthesized using the prepared CNTs as raw materials.

Herein, in this work, CNTs were *in-situ* deposited on the surface and in the core of the MgO castable *via* a chemical vapor deposition (CVD) process using PE as the primary material, and the effects of catalyst type, PE powder amount and pyrolysis temperature on the physical properties, high-temperature wettability to slag, and slag resistance of the prepared samples were examined. This study aims to utilize CNTs to enhance the slag corrosion resistance of MgO-based castables, thereby improving their service life in the steel industry.

2. Experimental

2.1. Raw materials

Magnesia grain and powder (MgO, ≥98 wt%, Youyan, LiaoNing, China) were used as raw materials for the preparation of the castable refractory. Waste PE powder with a size of 10–30 μm (≥97.0 wt%, Runwen material Co. Ltd., Shanghai, China) was used as the carbon source for CNT growth *via* the CVD method. Nickel nitrate, ferric nitrate and cobalt nitrate (Ni $(NO_3)_2·6H_2O$, Fe $(NO_3)_3·6H_2O$ and Co $(NO_3)_2·6H_2O$, AR, Sinopharm Chem. Co. Ltd., Shanghai, China) were used as precursors of the catalysts for the pyrolysis CVD reaction. Oleylamine (OAm, 80%-90 wt%, Bodi Chem. Co. Ltd., Tianjin, China) was used as a protective agent. CaO-Al_2O_3-SiO_2-MgO-Fe_2O_3-CaF_2 slag (CMASFF, Wuhan Iron and Steel Co. Ltd. Wuhan, China, and the chemical composition of the slag was shown in Table S1.) was used to test the high-temperature wetting behavior and corrosion resistance of the prepared CNTs/MgO castable.

2.2. Preparation of CNTs/MgO castable

The raw materials, proportioned according to Table S2, were firstly dry-mixed for 2–3 min. Next, 5.2 wt% water was added, followed by an additional 3 min of wet mixing. The resulting slurry was poured into molds (25×25×140 mm^3) under vibration to form castable samples. After curing at room temperature for 24 h, the samples were dried at 383 K for another 24 h to obtain the final MgO-based castable samples. The 1 wt%-CNTs/MgO castable was prepared *via* the CVD method using catalytic pyrolysis of PE powder based on the following reasons: 1) Further increasing the CNT content could cause its agglomeration, which would adversely affect the mechanical performance of as-prepared castable, and 2) the preparation of 2 wt% (or higher) CNTs/MgO castable requires more catalyst and a longer holding time, leading to higher costs.

Specifically, certain amounts of Ni $(NO_3)_2{\cdot}6H_2O$, Fe $(NO_3)_3{\cdot}6H_2O$ and Co $(NO_3)_2{\cdot}6H_2O$ were mixed with 50.0 mL ethanol to prepare catalyst precursor solutions with concentrations ranging from 0.1 to 1.5 mol/L. Subsequently, 1.0 wt% OAm was added to the precursor solution as a protective agent. The MgO-based castable samples with a size of 25×25×140 mm^3 were fully immersed in the aforementioned catalyst precursor solution, and vacuum-impregnated for 30 min. Finally, the dried castable samples were embedded in PE powder and subjected to heat treatment at 773–1173 K for 2 h under an argon atmosphere, with a heating rate of 5 K/min (Fig. 1). CNTs were *in-situ* deposited on the surface and in the inner part of the MgO castable using CH_4 and C_2H_4 generated from the decomposition of PE.

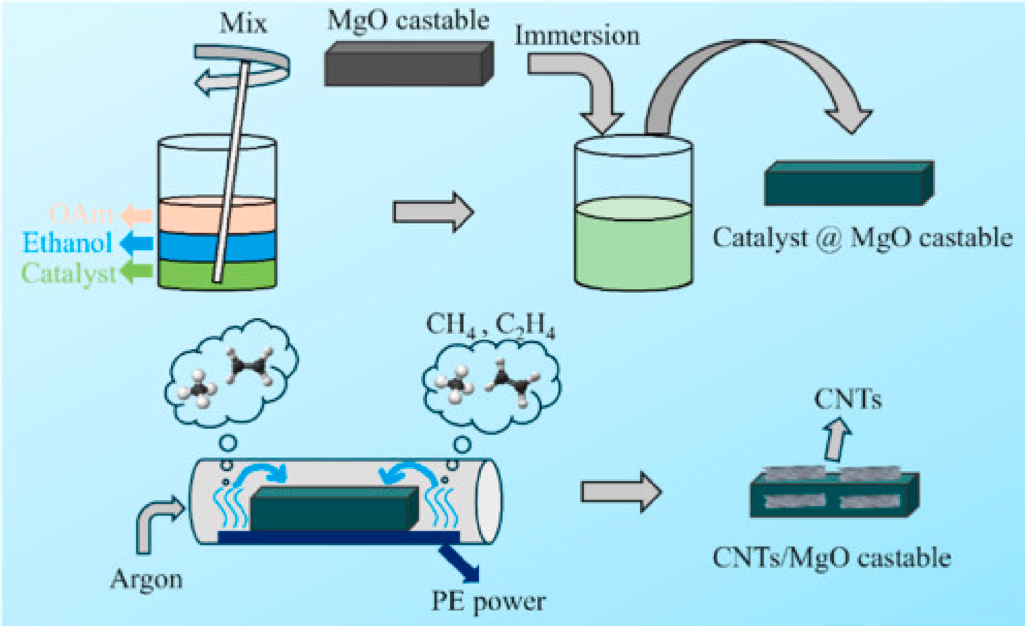

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Fig. 1. Preparation flowchart of CNTs/MgO castable.

2.3. Characterization

X-ray diffractometry was utilized to determine the crystalline phases present in the samples (XRD, X'Pert Pro, Philips, Netherlands) using Cu Kα radiation (λ=0.1542 nm). Raman spectral data were acquired using a Raman spectroscopy system (Horiba HR Evolution, UK) at a wavelength of 532 nm. FE-SEM (Quanta 400, FEI, USA, 15 kV) and high-resolution transmission electron microscope (FEI Talos F200x and JEOL JEM-F200) were used to characterize the microstructure and elemental composition of the prepared samples. Water contact angles were measured by an OCA15EC tester (DataPhysics, Germany). Bulk density and apparent porosity values were obtained through the Archimedes principle, and the cold compressive strength (CCS) was tested using a universal electronic test machine (EDC 120, DOLI Company, Germany, China). The morphological and microstructural characteristics of the samples were analyzed using an Apreo S Scanning Electron Microscope (Phenom, ProX, China) equipped with a 3D roughness reconstruction module.

The interfacial wetting characteristics of CMASFF on castable substrates at elevated temperatures were investigated using the sessile drop technique, with real-time monitoring of contact angle (θ), droplet diameter (d), and height (h) evolution (TA-Z16B01, Tianjin Zhonghuan Furnace Company, China). A cylindrical slag ingot (Φ5.5 mm×7 mm) was firstly put on the surface of the castable samples. The assembly was then purged with flowing argon and heated at a rate of 10 K/min to 1573 K, followed by further heating at 5 K/min to 1973 K and maintained at this temperature for 30 min.

3. Results and discussion

3.1. Effect of pyrolysis temperature on the preparation of CNTs/MgO castables

Fig. 2a presents the XRD patterns of the samples prepared at different pyrolysis temperatures (using 1.0 mol/L $Co(NO_3)_2{\cdot}6H_2O$ and the mass ratio of sample/PE was 1:1), indicating that only the diffraction peaks of MgO and Al_2O_3 were detected in the sample. Raman spectra of the prepared samples were then recorded, and the G/D intensity ratio (I_G/I_D) was commonly used to assess the graphitization of the formed CNTs, and a higher I_G/I_D ratio indicates a lower density of lattice defects in the CNT structure [[27], [28], [29]]. With increasing temperature from 773 K to 973 K, the I_G/I_D ratio of the samples increased from 0.93 to 1.01 (Fig. 2b), indicating that graphitization of CNTs was promoted at higher temperature. However, when the temperature further increased to 1173 K, the I_G/I_D ratio decreased to 0.84 (Fig. 2b), this could result from the reduced catalyst activity caused by the higher temperatures. The cross-sectional microstructure of the samples prepared at different temperatures was then characterized by SEM (Fig. 2c–g), and amount of CNTs were observed in the samples at all temperatures. With increasing temperature from 773 K to 1173 K, the carbon content increased from 1.4 wt% to 2.6 wt% correspondingly (Fig. 2h), indicating the successful preparation of a novel ultralow-carbon CNTs-modified MgO-C refractory.

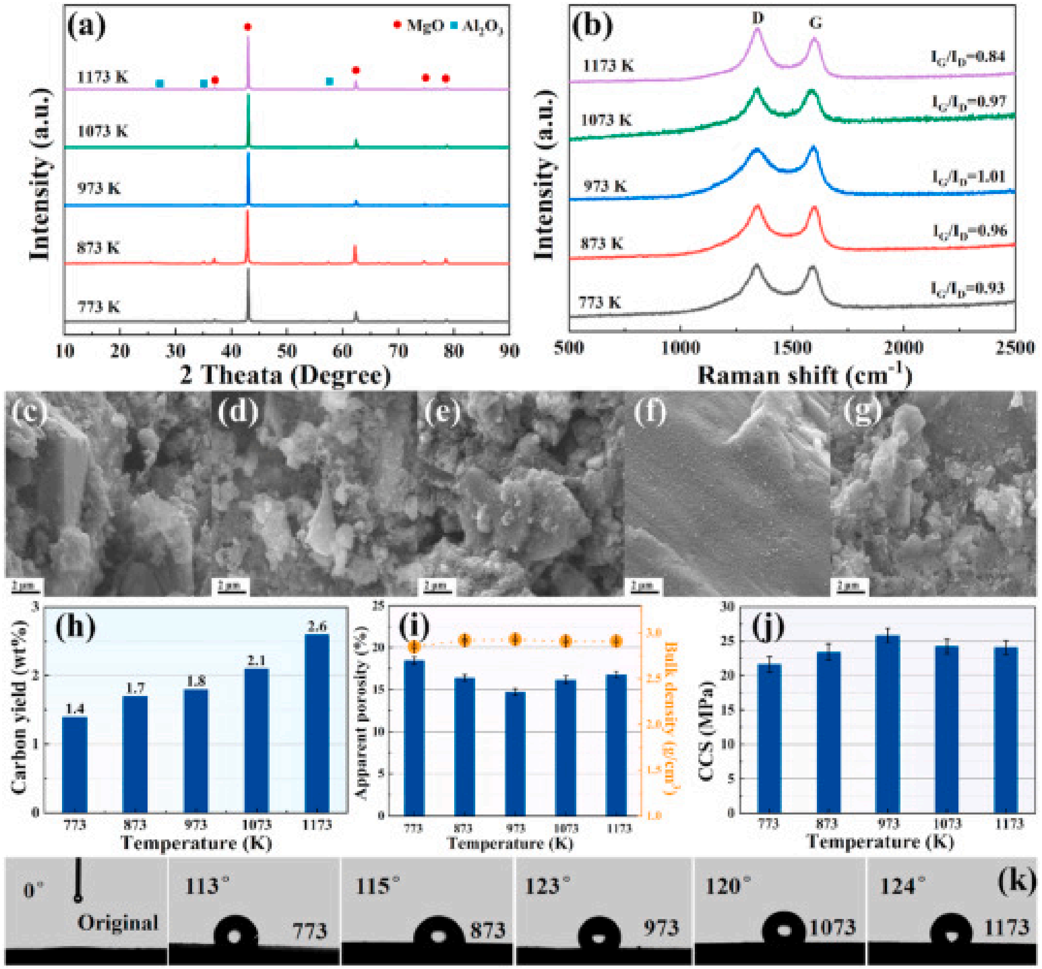

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Fig. 2. XRD patterns (a) and Raman spectra (b) of CNTs/MgO castables. SEM images of CNTs/MgO castables prepared at different pyrolysis temperatures: (c) 773K, (d) 873K, (e) 973K, (f) 1073K and (g) 1173K. (h) Carbon content of the CNTs/MgO castables pyrolyzed at different temperatures. Apparent porosity and bulk density (i) and CCS (j) of the sample pyrolyzed at different temperatures, and water contact angles (k) of the control MgO sample and the castables pyrolyzed at different temperatures.

Fig. 2i presents the room-temperature physical properties of the sample prepared with different pyrolysis temperatures, indicating that apparent porosity of the samples decreased whereas bulk density increased with the temperature increasing from 773 to 973 K. Further increase in temperature led to an increase in porosity, revealing that the optimal pyrolysis temperature in the present work was 973 K. Moreover, it is noticed in Fig. 2j that, the CCS of the prepared samples was consistent with the trend of bulk density, reaching a maximum value of 25.8 MPa at the pyrolysis temperature of 973 K. Fig. 2k presents the water contact angles of the samples after 2 h pyroly at 773–1173 K, demonstrating that the water contact angle of the samples increased with pyrolysis temperature. As the temperature rose to 973–1173 K, the water contact angles of the samples were about 120°. Compared with the water contact angle of 0° for the control MgO castable, a transformation from intrinsic hydrophilicity to hydrophobicity was obtained, which can be ascribed to the *in-situ* growth of CNTs on the surface of the sample.

3.2. Effects of catalyst concentration on the preparation of CNTs/MgO castables

Fig. 3a and b present the XRD and Raman spectra of samples fired at 973 K for 2 h with different Co $(NO_3)_2{\cdot}6H_2O$ concentrations. Similarly to Fig. 2a, only the diffraction peaks of MgO and Al_2O_3 were detected in the samples (Fig. 3a). The I_G/I_D ratios shown in Fig. 3b indicate that increasing the catalyst concentration promotes the graphitization of CNTs. The microstructure of the cross section of the samples shown in Fig. 3c–f indicated that CNTs with short-length were generated when the concentration of Co $(NO_3)_2{\cdot}6H_2O$ varied from 0.1 to 1.5 mol/L. In addition, when the catalyst concentration increased from 0.1 to 1.5 mol/L, the catalyst content of the samples increased from 0.32 to 1.6 wt% (Fig. 3g), and the corresponding carbon content increased from 1.3 to 2.3 wt% (Fig. 3g).

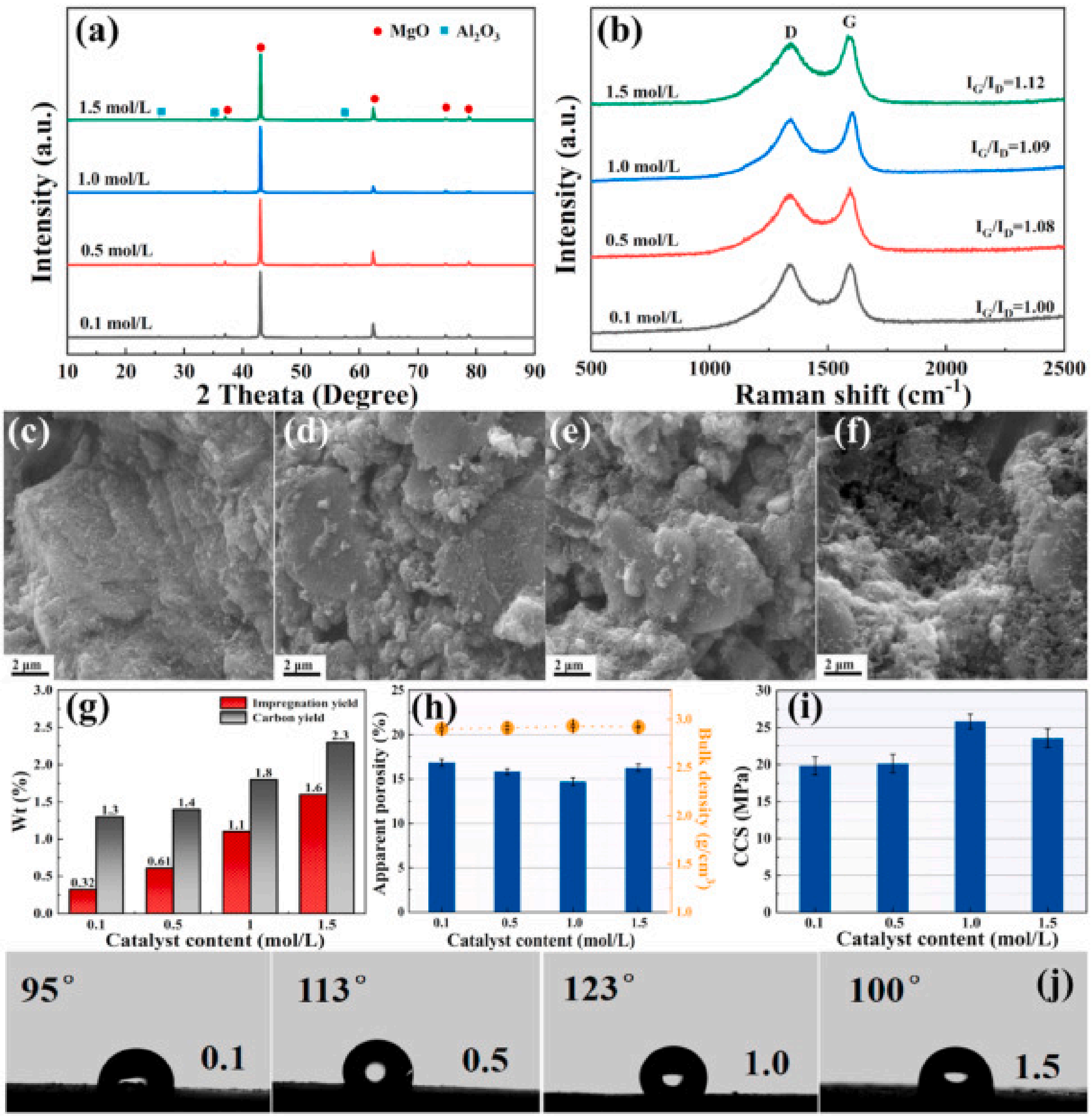

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Fig. 3. XRD (a) and Raman spectra (b) of CNTs/MgO castables. SEM images of CNTs/MgO castables prepared with different Co $(NO_3)_2{\cdot}6H_2O$ catalyst concentrations: (c) 0.1 mol/L, (d) 0.5 mol/L, (e) 1.0 mol/L and (f) 1.5 mol/L, (g) catalyst impregnation yield and carbon content of CNTs/MgO castables prepared with different Co $(NO_3)_2{\cdot}6H_2O$ catalyst concentrations. Apparent porosity and bulk density (h), CCS (i) and water contact angles (j) of the sample prepared with different Co $(NO_3)_2{\cdot}6H_2O$ catalyst concentrations.

Fig. 3h and i present the room-temperature physical properties of CNTs/MgO samples fabricated with different catalyst amounts. As shown in Fig. 3h, apparent porosity decreased whereas bulk density increased with the catalyst concentration increasing from 0.1 to 1.0 mol/L, which was due to the *in-situ* formed CNTs in the sample filling the pores. Upon further increasing the catalyst concentration to 1.5 mol/L, the porosity of the samples increased again, demonstrating that the excessive catalyst was not appropriate for the preparation of the CNTs/MgO refractory samples. As shown in Fig. 3i, the CCS trend was found to correlate with bulk density variations, reaching a maximum value of 25.8 MPa at 1.0 mol/L of catalyst concentration. Fig. 3j presents the surface water contact angle of the prepared samples, demonstrating that the water contact angle initially increased and then decreased with rising catalyst concentration, increasing within the 0.1–1 mol/L concentration range but decreased when the concentration further rose to 1.5 mol/L. The possible reason is that the increased catalyst concentration led to agglomeration of the catalyst, thereby deteriorating its catalytic performance.

3.3. Effect of the catalyst type on the preparation of CNTs/MgO castables

Fig. 4a shows the Raman spectra of CNTs/MgO castable fired at 973 K for 2 h using 1.0 mol/L Fe, Co or Ni as catalysts, demonstrating that the graphitization of CNTs reached the highest when Ni was employed as the catalyst. The order of catalytic activity for preparation of CNTs/MgO castable was Ni>Co>Fe. When using Ni as catalyst, the porosity of the prepared castable sample was the lowest and whose CCS reached a maximum value of 30.6 MPa (Fig. 4b and c). This can be attributed to the Ni catalyst promoting the growth of a large number of CNTs inside the samples (Fig. 4), filling the pores and thus improving the mechanical properties of the refractory sample.

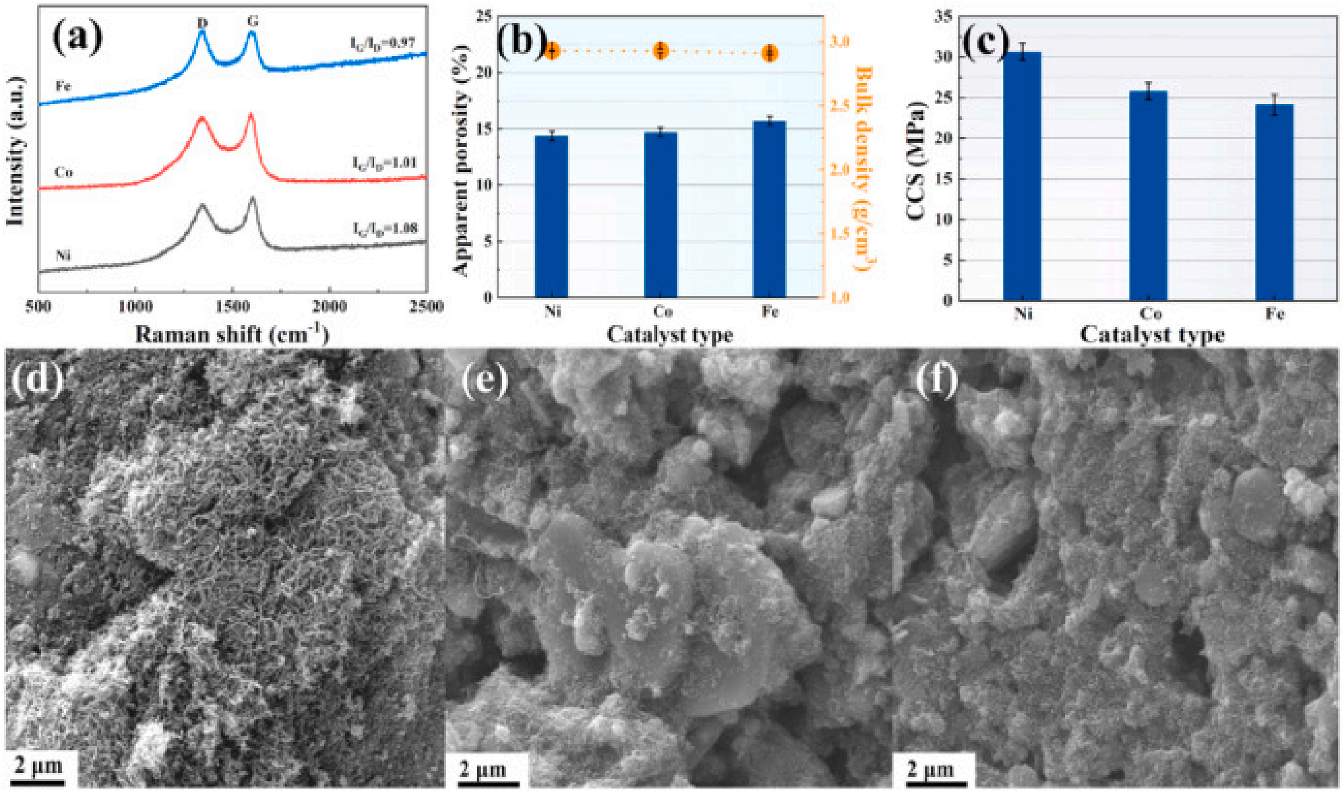

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Fig. 4. Raman spectra (a), apparent porosity and bulk density (b) and CCS (c) of CNTs/MgO castable samples prepared with different catalyst types. And SEM images of CNTs/MgO samples prepared with (d) Ni, (e) Co and (f) Fe as catalysts.

3.4. Effect of PE powder amount on the preparation of CNTs/MgO castables

The decomposition process of PE powder is shown in Fig. S1, confirming that complete decomposition occurs at 773 K, and that the main carbon-containing gaseous products are methane (CH_4) and ethylene (C_2H_4). Fig. 5a shows the XRD of CNTs/MgO castables prepared with different PE contents using Ni $(NO_3)_2{\cdot}6H_2O$ as catalyst, indicating that MgO and Al_2O_3 were observed in all samples. Fig. 5b presents the Raman spectra of CNTs/MgO castables prepared with different PE contents, demonstrating that the graphitization of CNTs increased with decreasing PE content. The reason may be attributed to the reduced dosage of PE powder, which allows the carbon-containing gases derived from PE pyrolysis to be more fully utilized, thereby resulting in a higher degree of graphitization. Fig. 5c–f show the cross-sectional SEM images of the prepared CNTs/MgO samples, revealing that a large amount of CNT grew in the case of a sample:PE mass ratio of 1:1 (Fig. 5c). The quantity and length of CNTs decreased as the PE content decreased. When the sample:PE mass ratio was 4:1, only a few worm-like CNTs were observed (Fig. 5f). This indicates that, although reducing the PE content enhanced graphitization degree, it significantly limited CNT formation, thereby inhibiting further CNT growth (Fig. 5b–f). As the sample/PE mass ratio increases from 1:1 to 4:1, the carbon content decreased from 2.2 wt% to 1.3 wt% (Fig. 5g).

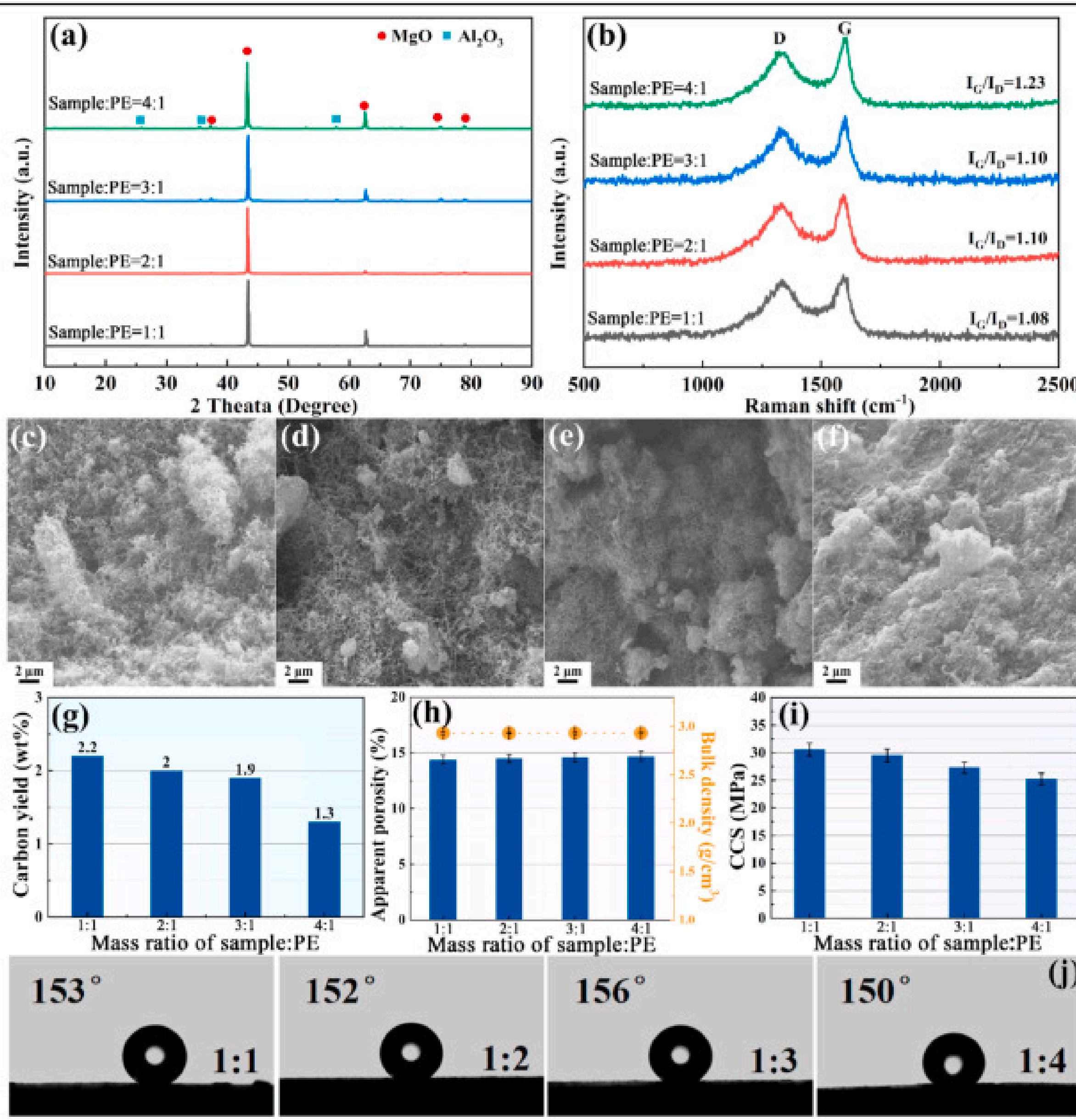

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Fig. 5. XRD patterns (a) and Raman spectra (b) of CNTs/MgO castables. SEM images of CNTs/MgO castables prepared with different sample/PE mass ratios: (c) 1/1, (d) 2/1, (e) 3/1 and (f) 4/1. (g) Carbon content of the CNTs/MgO castables prepared with different sample/PE ratios. Apparent porosity and bulk density (h), CCS (i) and water contact angles (j) of the samples prepared with different sample/PE ratios.

Fig. 5h and i show the room-temperature physical properties of the prepared CNTs/MgO castable. As shown in Fig. 5h, when the sample/PE mass ratio increased from 1:1 to 4:1, no significant changes were observed in either porosity or bulk density. The CCS reached its maximum value of 30.6MPa at the sample/PE ratio of 1:1 (Fig. 5i). The decrease in CCS with decreasing PE dosage may be due to the insufficient formation of CNTs, which therefore could not effectively fill the pores of the castable sample. Fig. 5j shows the surface water contact angles of the prepared CNTs/MgO castables, the water contact angle measurement revealed a value of 153° for the castable prepared with a 1:1 sample/PE mass ratio. It should be noted that the water contact angle of all the prepared castable samples remained consistently above 150°, and the superhydrophobic properties of the samples demonstrate the successful deposition of CNTs with reasonable high density.

3.5. High-temperature wettability of CNTs/MgO castable to slag

Based on the results shown in Fig. 2, Fig. 3, Fig. 4, Fig. 5, castable samples with different catalyst types and concentrations were prepared at a sample/PE mass ratio of 1:1 and a pyrolysis temperature of 973K, and their slag corrosion resistance was first evaluated. The preliminary slag corrosion resistance of these CNTs/MgO castables was evaluated at 1873K for 2h, and the corrosion morphologies and corrosion areas/(slag area) of the samples are shown in Figs. S2 and 6a. The CNTs/MgO sample prepared using Ni catalyst displayed the smallest corrosion area, demonstrating its superior slag corrosion resistance compared to other samples prepared with Co or Fe catalysts.

The high-temperature wettability to slag of 1wt%-CNTs/MgO castable sample ($Ni(NO_3)_2 \cdot 6H_2O$ as the catalyst, catalyst concentration of 1.0mol/L, pyrolysis temperature of 973K, sample:PE mass ratio of 1:1, the carbon content of the sample was 1wt%) was then tested using sessile drop method, and the softening and spreading behavior of the slag ingot were displayed in Fig. 6b. This revealed a morphological transition of the slag ingot from cubic to spherical crown shape with increasing temperature during the melting process. The wettability characteristics were qualitatively evaluated by analyzing droplet morphology after complete melting of the sample. Slag spreading on the 1wt%-CNTs/MgO castable sample was significantly slower, and a contact angle of approximately 50° was observed even at 2550s (Fig. 6b), indicating the improved slag resistance achieved by CNTs deposition on the MgO. The improvement in the high-temperature slag wetting angle is attributed to the deposition of CNTs on the castable sample. Moreover, owing to the one-dimension structure of the CNTs, the surface roughness of the modified castable sample also increased, resulting in large contact area between the molten slag and the CNTs/MgO sample, ultimately enhancing the slag resistance of the final product. In contrast, for the control MgO castable sample, a completely and rapidly wettability to the slag melt was observed under the same conditions [30]. For the commercial MgO-5 wt% C sample, during the initial testing stage (before 180s), the contact angle of the sample gradually increased. However, after this period, the contact angle of the sample dropped sharply to only 11°, which was much lower than that of 1 wt%-CNTs/MgO sample, as shown in Table S3 [30]. Fig. 6c shows the contact angle, height and diameter variation of molten slag on the 1 wt%-CNTs/MgO sample. The sample exhibited a substantially improved non-wetting property, achieving a maximum contact angle of 116° and maintaining non-wetting conditions for up to 424s. Moreover, from 870 to 1588s, both the slag height and diameter gradually increased, the slag height grew from an initial 4.6mm to a maximum of approximately 6.8mm. The volume expansion of slag was primarily caused by gas generation from the redox reaction between Fe_2O_3 and carbon (producing CO/CO_2), while subsequent high-temperature shrinkage resulted from gas release through the molten slag matrix. It should be emphasized that although the CNTs exhibited a clustered morphology (Fig. 4), the high-temperature non-wetting property of the sample was also improved.

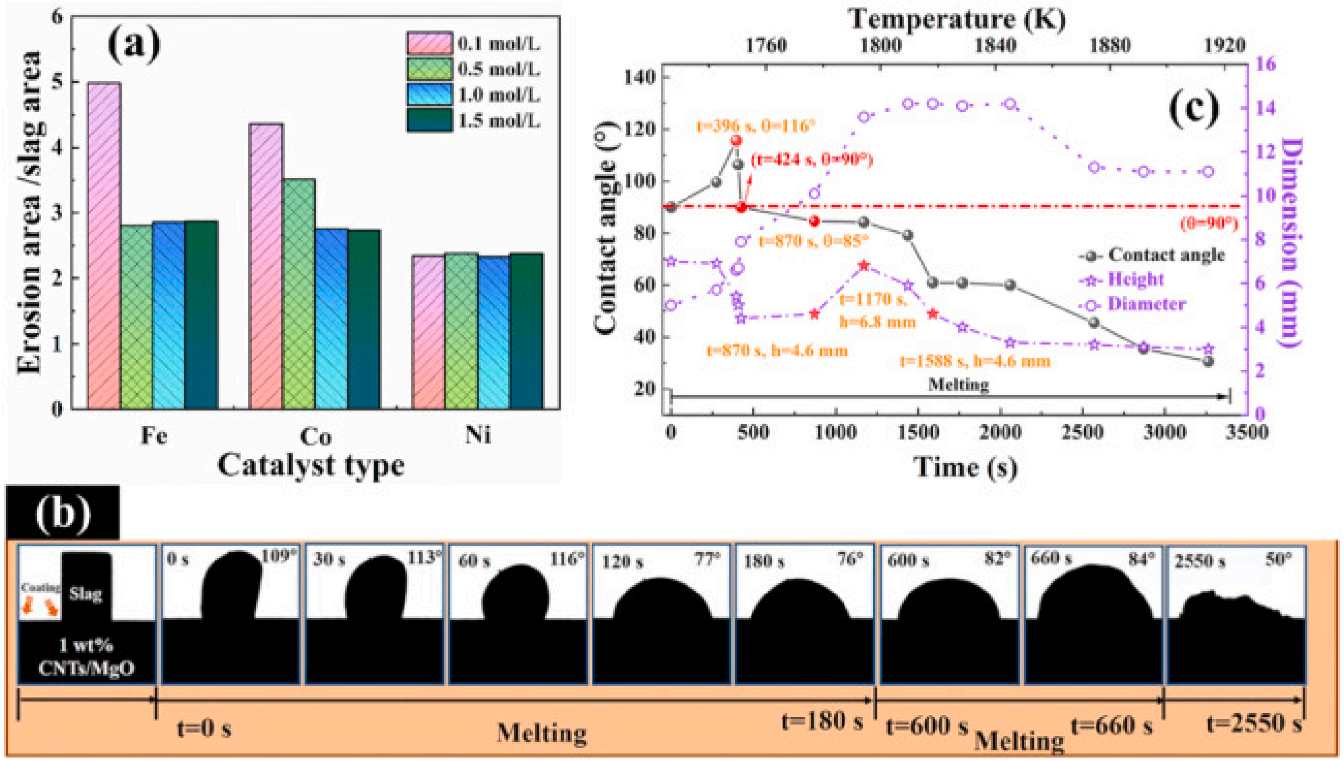

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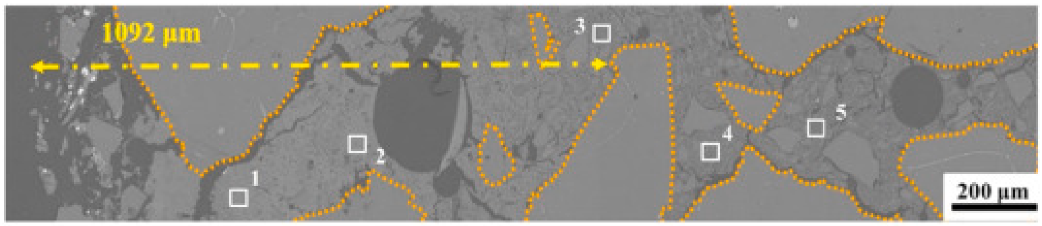
Fig. 6. (a) Slag corrosion on the surface of the castable samples. Optical images showing the morphology of slag droplets during the wetting test (b) and the wetting curves of molten slag (c) on the surface of the 1 wt%-CNTs/MgO sample surface.

Slag infiltration can be directly observed *via* the top view of the sample after the high-temperature wetting experiment, and the corresponding corrosion areas were then calculated. The slag completely penetrated into the control MgO sample, showing a black infiltrated circle with an area of about 235.4 mm^2 [30]. In contrast, a large amount of residual slag accumulates on the surface of the 1 wt%-CNTs/MgO castable sample, and thus a smaller corrosion area of about 217.1 mm^2 was exhibited (Fig. S3 and Table S4), which can be attributed to the non-wetting of the sample toward the slag.

In addition to the inherent non-wettability between carbonaceous materials and molten slag, the high roughness of the CNTs/MgO sample surface is another primary reason for the elevated high-temperature θ between slag and sample. The surface roughness of the samples was tested using the Apreo S SEM attached with a 3D roughness reconstruction module and shown in Table S4, indicating the much higher surface roughness of the 1 wt%-CNTs/MgO sample (82.9 nm *vs* 109.5 nm). To further investigate the penetration and corrosion of slag, cross-sectional microstructures of the tested sample were analyzed. Fig. 7 and S4 demonstrate that the 1 wt%-CNTs/MgO sample exhibited a corrosion depth of 1092 μm (based on the penetration depth of Ca and Si elements from the slag), which is only half of that control MgO sample (high up to 2308 μm [30]), confirming the excellent slag corrosion resistance of the CNTs modified sample. As shown in Fig. 7, after the slag corrosion test, numerous internal pores were formed in the original MgO castable [30], whereas the 1 wt% CNTs-MgO castable exhibited no evidently pore formation. This could be attributed to the CNT layer deposited on the sample surface, which effectively impedes slag penetration. However, it should be noted that, this study presently only assessed the high-temperature wetting angle performance of the prepared CNTs/MgO castable, the investigation of high-temperature mechanical properties of the sample is also very important for industrial application, and will be conducted subsequently.



EDS-Dot(at%)	Ca	Si	O	Mg	Al	Fe	C
1	8.06	6.51	41.05	12.52	20.20	0.13	-
2	5.06	8.79	30.01	37.12	7.52	0.32	-
3	1.89	1.27	38.11	35.7	11.53	11.5	-
4	0.52	0.45	48.11	15.32	34.78	0.82	-
5	0.12	0.24	45.1	18.5	35.85	0.19	-

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Fig. 7. BSE images and point-EDS mapping of 1 wt%-CNTs/MgO sample after the slag wetting experiment.

4. Conclusion

In this study, 1 wt%-CNTs/MgO castable was prepared by a PE pyrolysis CVD method, CNTs with non-wetting characteristics were *in-situ* grown on the surface and inside of the castable sample, which enhanced the bulk density and surface roughness of the final product. Ni $(NO_3)_2{\cdot}6H_2O$ exhibited better catalytic activity than Fe $(NO_3)_3{\cdot}9H_2O$ and Co $(NO_3)_2{\cdot}6H_2O$. The optimal conditions for preparation of 1 wt%-CNTs/MgO castable sample were as following: a catalyst concentration of 1.0 mol/L, a pyrolysis temperature of 973 K, using Ni $(NO_3)_2{\cdot}6H_2O$ as the catalyst, and a mass ratio of sample to PE of 1:1. The as-prepared 1 wt%-CNTs/MgO castable sample exhibited a bulk density of 2.93 g/cm^3, an apparent porosity of 14.4%, a compressive strength of 30.6 MPa, and a super-nonwetting ability to slag with a high-temperature contact angle of 116° and a nonwetting time of 424 s. The corrosion depth of 1 wt%-CNTs/MgO castable sample was only 1092 μm, which was less than half that of the control MgO castable sample. This CNT modification approach could be potentially applied to other types of castables, such as Al_2O_3, Al_2O_3-Mg Al_2O_4, and Al_2O_3-SiC-C castables.

CRediT authorship contribution statement

Chao Qu: Writing – original draft, Investigation, Formal analysis. **Xiaoyang Huang:** Validation. **Haijun Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Yage Li:** Writing – review & editing, Validation, Methodology. **Shaowei Zhang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is/are the supplementary data to this article:

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Multimedia component 1.

Recommended articles

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